High Availability (HA) System – Hot & Standby Notes

1. System Overview

- **Goal:** Ensure zero/minimal downtime by using **hot and standby Kubernetes clusters**.
- **Hot Cluster:** Active, serving traffic.
- **Standby Cluster:** Ready to take over if hot fails.
- **Traffic Management:** Global load balancer routes traffic to the hot by default, standby if failure.
- **Data & Config Sync:** Databases, object storage, and deployments replicated between clusters.

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2. Architecture Components

A. Kubernetes Clusters

- **Hot Cluster**
    - Multi-AZ for resilience.
    - Hosts all production workloads.
    - Stateful workloads use **replicated persistent volumes**.
- **Standby Cluster**
    - Mirrors hot cluster.
    - Sync deployments via Github Action CI + Fluxcd .

    - Can be hot-standby (near-live) or cold-standby (manual start).

B. Load Balancers

- **Global Load Balancer / DNS-based (Azure front door)**
    - Sends traffic to the hot cluster by default.
    - Fails over to the standby cluster if the hot fails.

- **Cluster-level Load Balancer**

    Fronts the Kubernetes cluster (opensource NGINX Ingress controller).

    - Routes internal traffic to services.

C. Network & Security

- Separate VPC/subnets per cluster.
- VPC peering or VPN between clusters.
- Security measures:
    - Kubernetes **NetworkPolicies** (pod-level access control)
    - TLS encryption for all traffic
    - RBAC for Kubernetes access
    - WAF (Azure front door)

D. Data Layer

- **Databases:** Multi-master or replicated (Postgres streaming, MySQL group replication)
- **Object storage:** Azure Blob replication
- **Sync:** CI/CD pipelines keep deployments in sync

E. Monitoring & Health Checks

- **One extra separate** cluster for **Prometheus + Grafana ,** monitor the hot and cold cluster
- Alertmanager in Prometheus + Azure Monitor

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3. Hot-Standby Traffic Flow

1. User → Global Load Balancer
2. Default: Hot Cluster → Services → Databases / Storage
3. If Hot Cluster fails → LB reroutes → Standby Cluster
4. Standby Cluster already has synced apps + replicated data → minimal downtime

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4. Best Practices

- **Kubernetes**
  - Multi-AZ clusters
  - Deployments for stateless apps, StatefulSets for DBs
  - Fluxcd HA (bootstrap on both hot and standby cluster, same git repo , different directory **/hot , **/standby)
- **Load Balancing**
  - Health checks at multiple layers
  - DNS failover for global HA(Azure front door health probe)
- **Data Management**

    - Async replication for heavy writes
    - Regular failover testing
- **Security**
    - TLS everywhere
    - Least-privilege RBAC
    - Audit logging
- **Monitoring**
    - Centralized logs (Loki Promtail + Grafana)
    - Alerts for failover events

Overall Architecture

```
Client
  │
DNS name
  │
Azure Front Door (Edge + WAF)
  ├── Health Probe → /status
Backend Pool
  ├──────────────── HOT Cluster (Primary)
  │          │
  │      Private Link(internal network)
  │          │
  │      Internal LoadBalancer
  │          │
  │      Kubernetes Ingress Controller
  │          │
```

```
│            Service
│            │
│            Pod
└──────────────── STANDBY Cluster (Failover) ( Same flow as hot , private link , internal lb……..pod)
```

K8s Cluster

In this assignment, **self setup k8s** used instead of cloud native k8s(like azure aks).

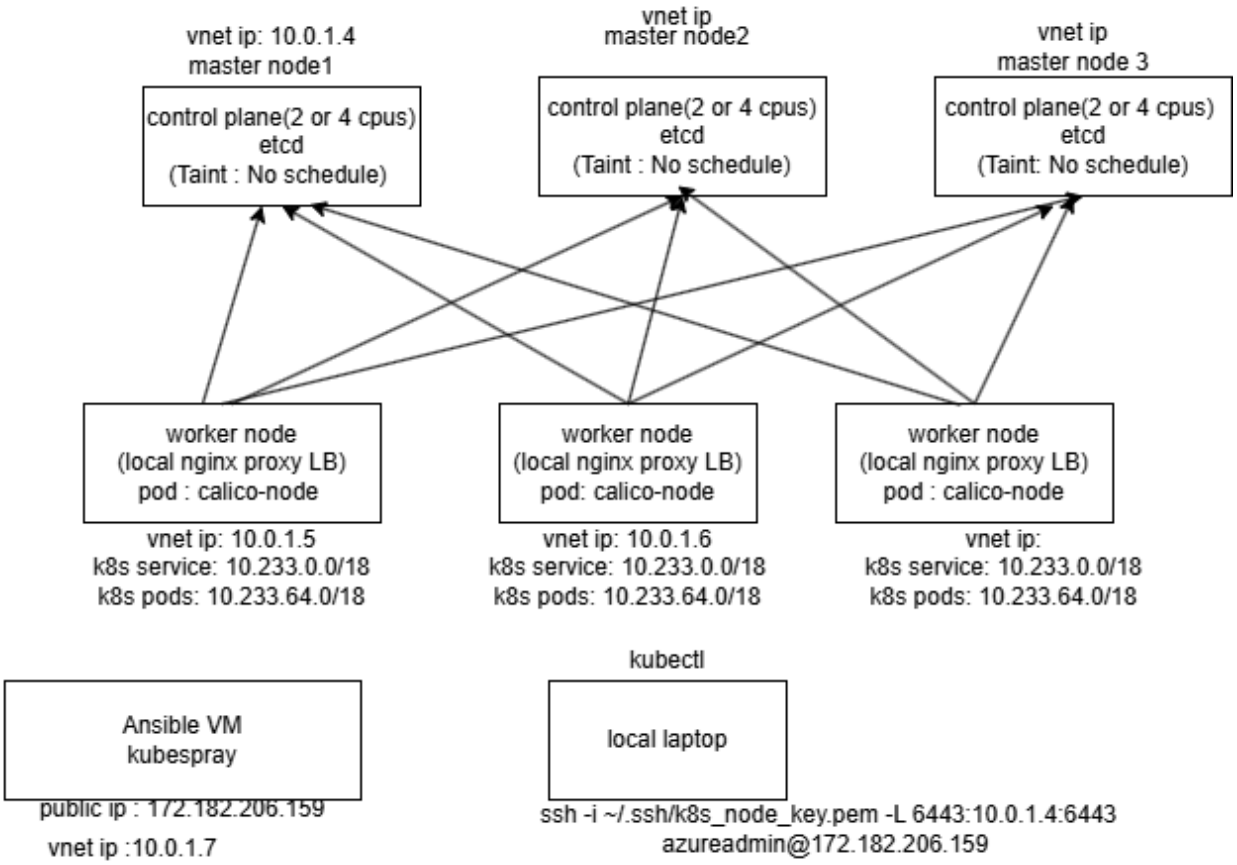
nodes : azure vm , generated by **terraform**
vnet: "10.0.0.0/16"
subnet: "10.0.1.0/24"
nsg:

Inbound Security Rules

100	AllowInternal	Any	Any	VirtualNetwork	VirtualNetwork	Allow
200	AllowSSHFromAdmin	22	Tcp	204.244.241.200	Any	Allow
210	AllowK8sAPI	6443	Tcp	204.244.241.200	Any	Allow

Ansible Automation based on opensource kubespray.

On_Prem K8S Architecture



For my lab , not all nodes are deployed as shown in the picture to save costs.

Install
ansible-playbook -i inventory/mycluster/inventory.ini -u azureadmin -b cluster.yml

Network config (CNI plugin , k8s service/pods cidr etc)
inventory/mycluster/group_vars/k8s_cluster/k8s-cluster.yml
kube_network_plugin: calico
kube_service_addresses: 10.233.0.0/18
kube_pods_subnet: 10.233.64.0/18

Upgrade : upgrade-cluster.yml

```
(.vnev) (base) bei@LAPTOP-LJ567OQ7:/mnt/c/Users/beibe/Downloads/BMO/backend-app$ kubectl get nodes -owide
NAME    STATUS   ROLES           AGE     VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE             KERNEL-VERSION      CONTAINER-RUN
TIME
node1   Ready    control-plane   4d19h   v1.35.1   10.0.1.4      <none>        Ubuntu 22.04.5 LTS   6.8.0-1044-azure    containerd://
2.2.1
node2   Ready    <none>          4d19h   v1.35.1   10.0.1.6      <none>        Ubuntu 22.04.5 LTS   6.8.0-1044-azure    containerd://
2.2.1
node3   Ready    <none>          4d19h   v1.35.1   10.0.1.5      <none>        Ubuntu 22.04.5 LTS   6.8.0-1044-azure    containerd://
2.2.1
node4   Ready    <none>          5h10m   v1.35.1   10.0.1.8      <none>        Ubuntu 24.04.3 LTS   6.14.0-1017-azure   containerd://
```

CICD pipeline

```
                CI Pipeline
                    │
                    │ build & push
                    ▼
            Azure Container Registry
                    │
                    │ (image tags)
                    ▼
             Flux Image Policy
                    │
                    │ update tag
                    ▼
                 Git Repo
            (ha-gitops-repo)
                    │
        ┌───────────┴───────────┐
        │                       │
        ▼                       ▼
   HOT Cluster             STANDBY Cluster
 (active traffic)         (passive replica)

Flux Controllers        Flux Controllers
├─ source-controller    ├─ source-controller
├─ kustomize-controller ├─ kustomize-controller
└─ image-automation-controller  └─ image-automation-controller

    │                       │
    ▼                       ▼
 apps/backend/hot        apps/backend/standby
    │                       │
    ▼                       ▼
 Kubernetes Deployment   Kubernetes Deployment
 replicas: 3             replicas:  3 / standby
```

```
(base) bei@LAPTOP-LJ567OQ7:/mnt/c/Users/beibe/Downloads/BMO/ha-gitops-repo$ tree
.
├── apps
│   └── backend
│       ├── base
│       │   ├── acr-secret.yaml
│       │   ├── backend-deployment.yml
│       │   └── kustomization.yaml
│       ├── hot
│       │   ├── backend-image-patch.yaml
│       │   └── kustomization.yaml
│       └── standby
│           └── kustomization.yaml
└── clusters
    └── hot
        └── flux-system
            ├── app-sync.yaml
            ├── gitrepository.yaml
            ├── gotk-components.yaml
            ├── gotk-sync.yaml
            ├── image-automation.yaml
            ├── image-policy.yaml
            └── kustomization.yaml
```

Github two repos ::

application: https://github.com/fanzhangjobs/backend-app (Dockerfile included)

Fluxcd: https://github.com/fanzhangjobs/ha-gitops-repo (k8s manifest included)

reusable base manifest ,Hot Environment Overlay (patch), Standby Environment Overlay (patch)

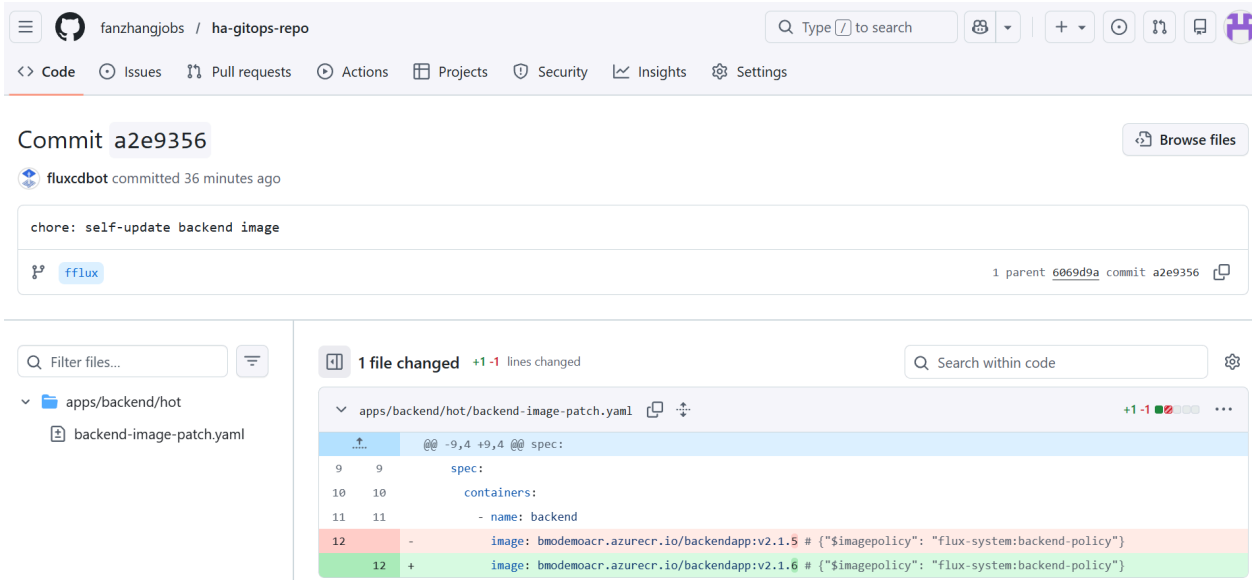
CI

application repo .github/workflows/build-and-push.yml

Run pytest ->build ->push to azure ACR ,image tag (support auto increment like 2.1.1 to 2.1.2 if no manual tag by github environment variable)

CD

Fluxcd (image update automation once new image pushed)



Note: For secrets.yaml we should encrypt it. Sops tool used . decryption needed in Kustomization in Fluxcd

Monitoring and logging

- Health probe in Azure Front Door (required)

- Prometheus + grafana

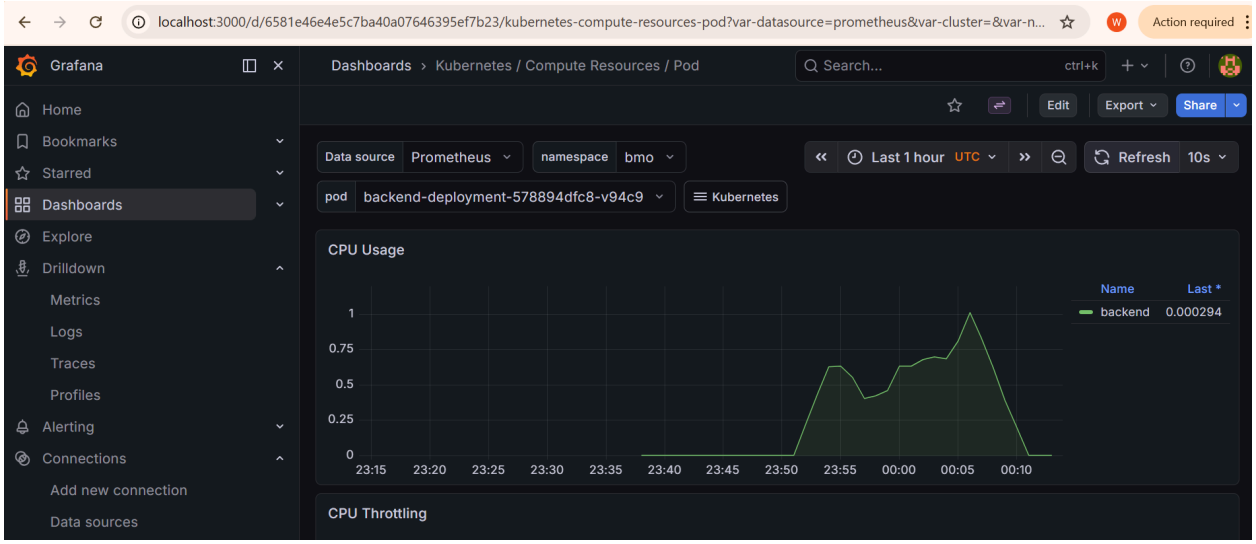
In my lab I didn't create an extra cluster to save cost. In production, extra clusters are needed.

Install helm chart prometheus-community/kube-prometheus-stack
grafana data source , connection , Prometheus server URL ,
http://prometheus-kube-prometheus-prometheus.monitoring.svc.cluster.local:9090

manually triggered cpu stress to monitor the cpu usage

```
(base) bei@LAPTOP-LJ567OQ7:/mnt/c/Users/beibe/Downloads/BMO/backend-app$ kubectl port-forward svc/backend-service -n bmo 4000:80
Forwarding from 127.0.0.1:4000 -> 5000
Forwarding from [::1]:4000 -> 5000
```

```
(.vnev) (base) bei@LAPTOP-LJ567OQ7:/mnt/c/Users/beibe/Downloads/BMO/prometheus$ curl -X POST http://localhost:4000/stress   -H "Content-Type: application/json"   -d '{"duration":300, "threads":20}'
{"status":"CPU stress started for 300s with 20 threads"}
```



- Shared central logging, Promtail → Loki → Grafana, Both clusters send logs to one Loki.

Todo:

- Set a standby cluster , configure azure front door to test the failover
- More complicated application(including db …)
- Test on k8s ha(master node failover)
- Network policy on k8s(now i only have one backend app in test)